Little's Formula

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Abstract

Time averages versus distribution means, Little's formula $\bar{N} = \lambda \bar{T}$ for $G/G/1$ in steady state, and its use on $M/M/1$ to split sojourn time into waiting and transmission.

1 Time and Steady State Averages

We have already introduced the concept of distribution means and sample based means. In this section we introduce a further type of average, a time based average. The time average is the average value of a particular function over time. If that function is continuous it is given by

$$\bar{f}_t = \frac{1}{t} \int_0^t f(\tau)\, d\tau.$$

The derivation of Little's formula that follows will rely on an assumption. We will assume that averages under the steady state distribution are equivalent to the time average in the limit as $t \to \infty$, where if the state of the system is given as a function of time as $N(t)$ the average number in the system as a function of time is given by

$$\bar{N}_t = \frac{1}{t} \int_0^t N(\tau)\, d\tau$$

and the time average after in the limit as $t \to \infty$ is given by

$$\bar{N} = \lim_{t \to \infty} \bar{N}(t).$$

which we assume is equivalent to $\langle k \rangle_{p(k)} \equiv \bar{N}$. See pg 156-157 in [Bertsekas and Gallager, 1992] for more details.

2 The Formula

Little's formula applies under steady state conditions (see Section **??**). Under the steady state assumptions it states,

$$\bar{N} = \lambda \bar{T}.$$

In practice we can see what this means by considering a simple example of a nightclub queue.

2.0.0.1 Question You are driving past a nightclub, wondering how long you will have to wait to go in. You notice on Thursday nights that the queue typically contains 20 people and that people arrive at a rate of 40 per hour. According to Little's formula, how long would you expect to have to wait to get in?

The waiting time is given by Little's formula. $\bar{N} = 20$, $\lambda = 40$ and $\bar{T} = \frac{\bar{N}}{\lambda} = \frac{20}{40} = \frac{1}{2}$hour.

2.1 Intuition

When phrased in these terms the formula seems fairly obvious. If we see there are twenty people in a queue and they enter a building at a rate of 40 per hour, we might expect to have to wait half-an-hour in the queue. In the examples section at the end of this chapter we build on this intuition with a simple graphical 'proof' of the formula based upon that given in [Bertsekas and Gallager, 1992]. The formula applies regardless of the arrival and service time distributions, it only relies on the system being in steady state. Little's formula can be used at the level of the entire system (queueing time and service time), as we have seen, or it may be applied at a lower level.

3 Average Delay in $M/M/1$

In the last chapter we computed that the average state of an $M/M/1$ queue was given by

$$\bar{N} = \frac{\lambda}{\mu - \lambda} = \frac{\rho}{1 - \rho}.$$

We can use Little's formula to tell us the average delay associated with the system. From $\bar{N} = \lambda \bar{T}$ we rearrange to recover the average delay as

$$\bar{T} = \frac{1}{\mu - \lambda}$$

This is not the time spent in the queue however, this is the time spent in the system which includes *propagation delay*, the delay in transmitting the packet.

3.1 Little's Formula for Queueing Time and Transmission Time

The total delay for packet switching systems can be divided into queueing time, W, and service time or transmission time, X. $T = X + W$. Similarly the average delay may be decomposed as

$$\bar{T} = \bar{X} + \bar{W}.$$

Furthermore, both the queue itself and the transmission may the be analysed independently using Little's formula.

3.2 The Transmission

If we assume that customers/packets do not leave the queue then the rate of arrivals at the queue must equal the rate of arrivals at transmission. If we define the the average number under transmission as ρ then we can write

$$\rho = \lambda \bar{X},$$

where $\bar{X}$ is the average time spend under transmission for each packet. For the $M/M/1$ queue the average transmission time is given by the mean of an exponential distribution, $\bar{X} = \frac{1}{\mu}$, so for the $M/M/1$ queue we have

$$\rho = \frac{\lambda}{\mu},$$

a formula which we have already used in the last chapter. The variable ρ is known as the utilisation factor. It represents the fraction of time that the transmission line (or server) is in use and is therefore a measure of efficiency. Note that it is also the steady state probability that the transmission line is in use. For a lossless system (one that doesn't drop packets) this probability is given by $1 - P(0)$. You can check that this is consistent for the $M/M/1$ by looking at equation (**??**). *Note*: time under transmission is often considered proportional to packet length. For packet length of L bits and bandwidth of C bits per second transmission time is $X = \frac{L}{C}$.

3.3 The Queue

Given λ, an arrival rate, $\bar{N}_Q$, the average number of packets waiting in the queue, and $\bar{W}$, the average packet waiting time, Little's formula gives

$$\bar{N}_Q = \lambda \bar{W}$$

for the number in the queue. For the $M/M/1$ queue $\bar{T} = \bar{X} + \bar{W} = \frac{1}{\mu} + \bar{W}$ so the average time spent waiting in the queue is given by

$$\bar{W} = \bar{T} - \frac{1}{\mu} = \frac{\rho}{\mu - \lambda}.$$

We can therefore recover the average number in the queue

$$\bar{N}_Q = \frac{\rho^2}{1 - \rho}.$$

3.4 Examples

Graphical Proof of Little's Formula. In this example we provide a graphical proof of Little's formula.

Simple example of Little's formula . You are at a queue in a bar and notice that the bar tender spends 10% of her time cleaning between serving customers. If the typical number of people waiting to be served is 3 , what is the ratio of time spent waiting to being served?

3.5 Thanks!

For more information on these subjects and more you might want to check the following resources.

- company: Trent AI
- book: The Atomic Human
- twitter: @lawrennd
- podcast: The Talking Machines
- newspaper: Guardian Profile Page
- blog: http://inverseprobability.com

3.6 Further Reading

- Little's theorem and the $M/M/1$ delay formulae of Bertsekas and Gallager [1992]

3.7 Exercises

- Lab 1: sample-mean occupancy versus $\bar{N} = \rho/(1-\rho)$, and when the time average is not yet the steady state of Lawrence [2006]

References

Dimitri P. Bertsekas and Robert G. Gallager. *Data Networks*. Prentice-Hall, Upper Saddle River, NJ, second edition, 1992. ISBN 0-132-00916-1.

Neil D. Lawrence. Discrete simulation of the $M/M/1$ queue. Network Performance Analysis Lab Sheet 1, University of Sheffield, 2006. URL https://mlatcl.github.io/npa/.